AI: Strategy + Marketing (MGT 853)

AI in Retail + Fashion (Session 9)

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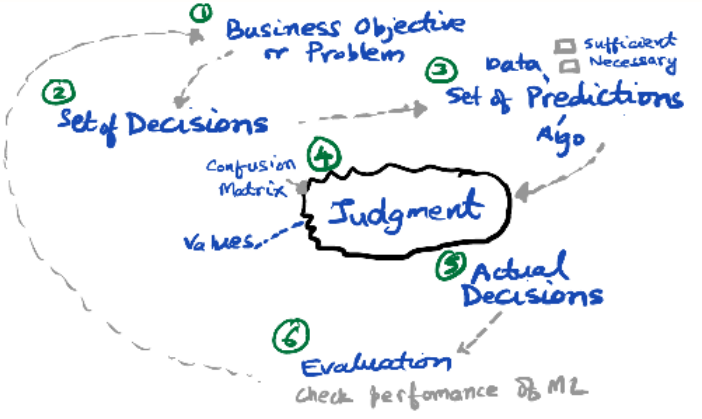
Course Logistics

1. Guest Speakers – Please sign up for lunch (need accurate count)
2. Project feedback ?
3. All assignments are now complete

Framework for ML in Business Problems



Challenges in the Fashion Retail industry

Miroglio in Fashion Retail

- Differences across store
- Proliferation of SKU
- Short
  - lifecycle
  - shelf life
- Demand Uncertainty
  - Tastes & Trends
  - Stockouts
- Production lead time
- Feedback time longer

M in competitive landscape

"Fast Fashion"

Competitive actions impact M much more

Business Problems $\Longrightarrow$ Decisions

(M)'s Business Problem / Objective:

Demand Uncertainty

DECISIONS:

1. Centralized / Distributed decisions
2. Inventory allocation to stores (SKU size)
3. Initial (85%) / Replenishment
   ↳ store product?
4. Promotion / Markdown
5. Order at beginning of Season (SKU-size)

Better Forecast
More data-driven
Efficiency

Metrics
- Inventory turnover ↑
- Stockouts ↓
- Cost of Logistics

Non-ML approaches

Rules of Thumb:

1. District managers reallocate SKUs
2. Sales in Past + Δ
3. Linear Forecasting
4. Decentralize.

Some non-ML approaches to solving the business problems

- ⇒ Central inventory pool for online
- ⇒ Proportion of customers entering/buying
- ⇒ Customer survey for out of stock

- → Human decisions
- → Benchmark competition
- → Historical Data (manager)
- → EOQ (Economic order qty)
- → Simplify by using a rule of thumb

- Understand human processes & current state of
- AI is a tool so need to benchmark performance

Decisions $\Longrightarrow$ ML Problems

- y^P: predicted sales quantity, y^O: order quantity
- Should we predict at SKU-size level or SKU-size-store level?
- How does the above depend on *which business decision* you are making?
- Predicted Quantity is a predcition but Order Quantity is a decision
- Important to be clear about the difference between Prediction and Decision
- What are some reasons that $y^O \neq y^P$?

Convert DECISIONS to PREDICTION Problems

Sales Quantity of SKU-size = Historic data, Trends, Ratings, Competitor (Macro), Product-level details, Price

$y^P = f(X)$

Where do we get data?

First Third

Decision on Order Quantity

Even with 100% accurate prediction, order Q $\neq$ Predicted Sales Q

Should Order Q = Predicted Sales Q?

y^P = Predicted Sales Q PREDICTION

y^O = Order Q , DECISION

Should $y^O = y^P$? 100% accuracy

Yes: ✓

No : Lost/Damage, (+)
Customer Perception (+)
Scarcity (−) → cycle of visits
immediate purchase
Uncertainty (+)

$y^O \neq y^P \Rightarrow$ Business Decision

y = Sales Q
per SKU
per Store
per size

$\simeq$ 1−2 / year
(sparsity in y)

Using Prediction in Decision Making

Equivalent of Confusion Matrix when y is continuous

- Do we observe all the data required to evaluate the loss?
- No, y^A is not observed when $y^P < y^A$ (stockout)
- Who might have information about excess demand that was stocked out?
- What information could they have?

y^P : Predicted Q,
y^A : Actual Q.

$y^P > y^A$

Loss : $(y^P - y^A) \times$ Cost of Salvage

$y^P < y^A$ + $(y^A - y^P) \times$ Cost of stockout

Brand ↓

We do not observe stocked out customers!

↳ Store Manager

How can Human help in AI decision making

Human $\Longleftrightarrow$ AI

How to get store managers input?

- Human has knowledge based on customer interaction

Test & Learn

Input to AI model:

X : Conversation, Local knowledge trends, preferences, group identity, Satisfaction on avail.

AI model with Store manager input

In class Exercise

Small Group Discussion

Forms groups of 3 or so, and explore the following questions.

- How would you design an AI model that would include store manager input?
- Detail the form of the manager input. Do you want X or Y variables? or both?
- How would you validate that the input is correct?
- How would you think about incentives for the manager to share this information?

Obtaining and Using Manager Input

How to use Store Manager Input?

- Estimation of sales as X
- Provide prediction y^p and allow to adjust - ± 10%
- market sentiment
- Prediction of need
  ↳ track manager predictions
- ↳ Compensation
- Qualitative reflection

Knowledge of customers sizes & preferences
selling process
conversations
Local trends

Miroglio – What Happened

- After much internal debate, Francesco Cavarero and Davide Garelli decided to partner with Evo Pricing (Evo) to build an artificial intelligence (AI) system for demand forecasting and inventory replenishment for Miroglio's Elena Mirò brand.
- Evo CEO Fabrizio Fantini and his team of data scientists worked in close collaboration with Garelli's team and with Miroglio's Chief Data Officer, to come up with the following system.

Miroglio – Designing AI system - A

Initial Inventory Allocation Decisions:

1. First decision depended on the accuracy of and confidence in the new replenishment system. Based on the system described below, Evo decided to allocate 60% (↓ from 85%) of the total inventory at the start of the season.
2. Second followed Miroglio's segmentation of store clusters, ensuring that its larger stores received more inventory of the same assortment.
3. Third decision of allocating specific items to a store was done by first identifying features of each item (e.g., its fabric, category, color, price point etc.). Item matched to previous items with similar features sold in store.
4. Finally, the model computed the initial allocation of the item as the weighted first four weeks' sales volume of those other items.
5. Expensive black blouse was more likely to be allocated to stores that sold many black items (not just blouses), had higher sales of blouses (of all colors), and sold more expensive items

Miroglio – Designing AI system - B

Inventory Replenishment Decisions:

1. Fantini and his team knew that individual Miroglio stores sold few units in each size of SKU $\implies$ data at disaggregated level too sparse to predict accurately
2. Aggregated past sales of an item in a given category (e.g., women's pants) across all sizes to first build a forecast at an item-store level
3. Removed the effects of markdowns on past sales and estimated a seasonality coefficient. Then adjust item-store forecasts (How?)
4. Finally, they arrived at the size allocation for each item-store forecast by giving equal weight to two factors: relative frequency of a certain size for an item across all stores; and relative frequency of a size for a store across all items
5. Leveraging Image Data: Analyze images of fashion items to abstract key attributes (these might include not only attributes typically identified by humans, such as shape and design, but also many other unique aspects that might be hard for people to describe). It would then correlate these image attributes with sales.

Miroglio – Designing AI system - C

Including Manager Input on Customers and Local Trends:

1. Idea: providing each store manager with a **non-monetary budget** that they could use to accept or replace replenishment items that Miroglio's HQ proposed.
2. Garelli explained: "Our Head of Buying has a budget that she uses to decide how many blue jeans or black blouses to buy based on a demand forecast. However, she uses her judgment to modify these quantities from the demand forecast. We can offer a similar budget to store managers and effectively make them the buying heads of their own stores."
3. Head of Retail Carlo Tibaldi was very excited about this idea: approach would empower store managers and encourage them to embrace the analytical model.

Budget

Budget = store's potential sales in the next four weeks + safety stock of 20% - current stock

Takeaways - 1

- Fashion is a challenging market to make predictions for several reasons. These factors are less relevant for some industries like groceries.
- Fast moving $\implies$ more frequent data $\implies$ better ML performance
  - But, if data is sparse, then hard to solve with ML.
- Specify business problems by formulating them as decisions
  - Check if non-ML approaches can be used (benchmark)
  - Convert to ML (prediction) problems.
  - Source of data: Unstructured (images) may be useful here.
- Challenge here is that important data may NOT be available to ML system

Takeaways - 2

- Human may have access to that data, likely unstructred based on conversations
  - Challenging to capture that data because human needs to be incentivized
  - Verifying human input may not be possible.
  - Need to understand and design incentives carefully. Build in flexibility for humans to leverage their local knowledge.

Bottomline

AI challenges are embedded in a larger socio-economic context. Need to understand the ecosystem and design everything around the AI system.